Pricing Total Return Swap

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Setting and Goal

A Total Return Swap (TRS) consists of two legs between a bank B and a customer C:

- **Security leg.** Pays the total return of a reference security S_t (price return plus dividends/coupons) over each period $(t_{i-1}, t_i]$.
- **Funding (premium) leg.** Pays $M_0\, r_f\, (t_i - t_{i-1})$ where $r_f = \ell + s_f$ is a floating rate index ℓ (e.g. 3M Libor) plus fixed spread s_f, and M_0 is the funding notional (typically $M_0 = S_0$).

The paper resolves the so-called *TRS pricing dilemma*: classical Libor discounting forces the funding premium to be paid up-front via day-1 CSA collateral call, while replacing it with a firm-specific funding rate violates the law of one price. The fix is to push the hedge financing cost onto the *security leg* and keep the TRS-as-a-whole discounted at the rate dictated by its collateralization scheme.

Cast of rates.

- r – riskfree / OIS short rate; $D(t,T) = \mathbb{E}_t^Q[e^{-\int_t^T r\,du}]$.
- $r_s \geq r$ – repo rate at which the hedge stock/bond is financed.
- r_b, r_c – senior unsecured rates of B, C; $r_w = r_b \mathbf{1}\{W \leq 0\} + r_c \mathbf{1}\{W > 0\}$.
- $r_N \geq r$ – B's rolling debt rate (matches r_b in the standard setup).
- λ – bond default intensity (Cox process); $q(t,T) = \mathbb{E}_t^Q[e^{-\int_t^T \lambda du}]$.
- $\mu_t = L_t/V_t \in [0,1]$ – collateralization ratio. Full CSA: $\mu \equiv 1$. Uncollateralized: $\mu \equiv 0$. Repo-style: see (9).

Wealth split. Exposure is split as $V_t = L_t + W_t$ where L_t is cash collateral and $W_t = V_t - L_t$ the residual unsecured amount carried at r_w.

1 Standard pre-crisis valuation

For a one-period TRS with $M_0 = S_0$ and continuous dividend yield q, the payoff to the payer (B) is

$$H(T) \;=\; M_0\, r_f\, (T - t_0) \;-\; \big(e^{\int_t^T q\,du}\, S_T - S_0\big). \tag{1}$$

Risk-neutral valuation under risk-free financing of the stock gives

$$V(t) \;=\; \mathbb{E}_t^Q\Big[e^{-\int_t^T r\,du}\, H(T)\Big] \;=\; \big(M_0\, r_f(T - t_0) + S_0\big) D(t,T) \;-\; S_t. \tag{2}$$

For the at-the-issue (ATI) case with $r_f = \ell + s_f$ and $(1 + \ell(T - t_0))D(t_0, T) = 1$, this collapses to

$$V_0 \;=\; s_f(T - t_0)\, S_0\, D(t_0, T),$$

i.e. the present value of the funding spread, which the borrower must post on day-1 under a CSA – the *repo trader's complaint.*

2 TRS hedge financing cost (full CSA)

The bank delta-hedges by holding $\Delta_t = -\partial V/\partial S$ shares financed in the repo market via account $L_t^s = \Delta_t S_t$ at rate r_s. The wealth equation pre-default is

$$\pi_t \;=\; M_t + (1-\Gamma_t)\big(V_t - L_t - W_t - N_t + \Delta_t S_t - L_t^s\big), \tag{3}$$

with $\Gamma_t = \mathbf{1}\{\tau \le t\}$ the joint default indicator. The self-financing condition reads

$$\begin{aligned} dM_t \;&= rM_t\,dt + (1-\Gamma_t)\big[\,dW_t - r_w W_t\,dt + dN_t - r_N N_t\,dt + dL_t - rL_t\,dt \\ &\quad + \Delta_t S_t q\,dt - d\Delta_t(S_t + dS_t) + dL_t^s - r_s L_t^s\,dt\big]. \end{aligned} \tag{4}$$

Setting $M_0 = N_0 = \pi_0 = 0$, demanding $\pi_t \equiv 0$, and applying Itô to $V(t,S)$ with $dS = \mu S\,dt + \sigma S\,dZ$ produces the pricing PDE

$$\boxed{\frac{\partial V}{\partial t} + (r_s - q)S\frac{\partial V}{\partial S} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - \mu r\,V - (1-\mu)r_w\,V \;=\; 0} \tag{5}$$

with $r_e(t) = \mu r + (1-\mu)r_w$ as the *effective discount rate*. Feynman–Kac gives

$$V(t) \;=\; \mathbb{E}_t^Q\Big[e^{-\int_t^T r_e\,du}\,H(T)\Big]. \tag{6}$$

Full collateral closed form. With $\mu = 1$ ($L_t = V_t$, $W_t = 0$): under Q, S has drift $r_s - q$, hence $\mathbb{E}_t^Q[e^{-\int_t^T r\,du} S_T] = S_t\,e^{\int_t^T (r_s - r)du}$, which gives the central closed-form

$$\boxed{V(t) \;=\; \big(M_0\,r_f(T-t_0) + S_0\big)D(t,T) \;-\; S_t\,e^{\int_t^T (r_s - r)du}} \tag{7}$$

The OIS-discounted funding leg is the *first* term; the asset leg becomes S_t scaled by the repo-vs-OIS funding factor.

The fva. Comparing (2) (which assumes $r_s = r$) with (7), the value reduction is

$$\boxed{fva \;=\; \big(e^{\int_t^T (r_s - r)du} - 1\big)\,S_t \;\ge 0} \tag{8}$$

This is the present value of repo-vs-OIS spread on the hedge – a *hedge financing* cost adjustment, distinguished from the upper-case "FVA" associated with uncollateralized swaps.

Equity forward consistency. Set $r_f = 0$, $M_0 = 0$, $S_0 \to F$, $H(T) = S_T - F$. Using (7): $V(t) = 0$ gives $F = S_t\,e^{\int_t^T (r_s - q)du}$, the standard cost-of-carry forward at the repo rate.

3 Repo-style margined TRS

Under repo-style margining, the variation margin posted by C is calculated like a repo balance:

$$L_t \;=\; S_0 - S_t + r_F\,M_0\,(t - t_0), \tag{9}$$

where r_F is the agreed margin accrual rate (typically the funding leg fixed leg). PDE (5) still holds with L_t specified above and $W_t = V_t - L_t$ (which can be negative – over-collateralized – without breaking anything).

Symmetric-funding closed form ($r_b = r_c \Rightarrow r_w = r_b$). For a one-period TRS, Feynman–Kac yields the integral form

$$V(t) = \mathbb{E}_t^Q\left[(M_0 r_f(T-t_0) + S_0 - S_T)e^{-\int_t^T r_b du} + \int_t^T e^{-\int_t^u r_b ds}(r_b - r)(S_0 - S_u + r_F M_0(u-t_0))du\right], \tag{10}$$

which integrates explicitly (deterministic rates, $D_{T,x} = e^{-\int_t^T x du}$):

$$\begin{aligned} V(t) = -S\left(\frac{r_b - r - (r_s - r)D_{T,r_b - r_s}}{r_b - r_s}\right) &+ \left(S_0 - t_0 M_0 r_f\right)\left(1 - r\,\frac{1 - D_{T,r_b}}{r_b}\right) \\ &+ M_0 r_f\left[T D_{T,r_b}\frac{r}{r_b} + \left(t + \frac{1 - D_{T,r_b}}{r_b}\right)\frac{r_b - r}{r_b}\right]. \end{aligned} \tag{11}$$

For asymmetric funding ($r_b \neq r_c$) the solution is no longer analytic; use the tree.

4 Trinomial tree (Boyle / repo-rate drift)

Build the tree on the asset S with up/middle/down factors

$$u = e^{\sigma\sqrt{2\Delta t}}, \qquad m = 1, \qquad d = 1/u,$$

and probabilities derived to match drift $r_s - q$ and variance σ^2:

$$p_u = \left(\frac{e^{(r_s - q)\Delta t/2} - e^{-\sigma\sqrt{\Delta t/2}}}{e^{\sigma\sqrt{\Delta t/2}} - e^{-\sigma\sqrt{\Delta t/2}}}\right)^2, \quad p_d = \left(\frac{e^{\sigma\sqrt{\Delta t/2}} - e^{(r_s - q)\Delta t/2}}{e^{\sigma\sqrt{\Delta t/2}} - e^{-\sigma\sqrt{\Delta t/2}}}\right)^2, \quad p_m = 1 - p_u - p_d. \tag{12}$$

The discount rate at each node is the switching r_e from (5); the sign of $W_t = V_t - L_t$ is fixed at the *previous* time step to decouple the nonlinearity (Lou 2017).

Multi-period reset (recursive scheme). Reset dates $\tau_0 < \tau_1 < \cdots < \tau_K = T$; previous fixing $S_{\tau_{i-1}}$:

1. Roll back from T to τ_K with payoff $M_{K-1} r_{f,K-1}(\tau_K - \tau_{K-1}) - (S_{\tau_K} - S_{\tau_{K-1}})$.
2. At each reset τ_i, for each node j with fixing $S_{i,j}$, *recurse* from that node forward to value the remaining strip; add the period return $S_{i,j} - S_{\tau_{i-1}}$; roll back to τ_{i-1} as in step 1.
3. Continue until reaching t_0.

5 Uncollateralized TRS and CRA decomposition

Let V^* be the OIS-discounted (counterparty-risk-free) TRS value and $U = V^* - V$ the total CRA. Then U solves

$$\frac{\partial U}{\partial t} + (r_s - q)S\frac{\partial U}{\partial S} + \tfrac{1}{2}\sigma^2 S^2\frac{\partial^2 U}{\partial S^2} - r_w U - (r_w - r)(V^* - L) \;=\; 0, \quad U_T = 0, \tag{13}$$

with closed form

$$U(t) \;=\; \mathbb{E}_t^Q\left[\int_t^T (r_w - r)\big(V^*(s) - L(s)\big)e^{-\int_t^s r_w du}\,ds\right]. \tag{14}$$

Spread decomposition. Decompose $r_b - r = s_b + \mu_b$ (CDS premium + bond/CDS basis), and analogously for C. Then $U = cva - dva + cfa - dfa$ with

$$cva(t) = \mathbb{E}_t^Q\left[\int_t^T s_c\,\mathbf{1}\{V(s) > 0\}\,(V^*(s) - L(s))\,e^{-\int_t^s r_w du}\,ds\right], \tag{15}$$

$$dva(t) = -\mathbb{E}_t^Q\left[\int_t^T s_b\,\mathbf{1}\{V(s) \le 0\}\,(V^*(s) - L(s))\,e^{-\int_t^s r_w du}\,ds\right], \tag{16}$$

$$cfa(t) = \mathbb{E}_t^Q\left[\int_t^T \mu_c\,\mathbf{1}\{V(s) > 0\}\,(V^*(s) - L(s))\,e^{-\int_t^s r_w du}\,ds\right], \tag{17}$$

$$dfa(t) = -\mathbb{E}_t^Q\left[\int_t^T \mu_b\,\mathbf{1}\{V(s) \le 0\}\,(V^*(s) - L(s))\,e^{-\int_t^s r_w du}\,ds\right]. \tag{18}$$

The discount rate is the *liability-side* senior unsecured rate (r_w). B's cva is exactly C's dva by construction – law of one price preserved.

6 TRS on defaultable bond

Replace stock S by bond B_t priced under default intensity λ, repo'd with haircut h: $L_t^b = (1-h)B_t$ at rate r_s, the remaining hB_t funded at r_N. The all-in bond financing rate is

$$\bar{r}_s \;=\; (1-h)r_s + h\,r_N. \tag{19}$$

Risky bond PDE. With $d\lambda = a\,dt + b\,dZ$,

$$\frac{\partial B}{\partial t} + a\frac{\partial B}{\partial \lambda} + \tfrac{1}{2}b^2\frac{\partial^2 B}{\partial \lambda^2} - (\bar{r}_s + \lambda)B + \lambda R \;=\; 0, \tag{20}$$

giving

$$B(t) = \mathbb{E}_t^Q\left[e^{-\int_t^{T_b}(\bar{r}_s+\lambda)du} + \int_t^{T_b}\lambda R\,e^{-\int_t^{\tau}(\bar{r}_s+\lambda)du}d\tau + \sum_{i:T_i>t} c_i e^{-\int_t^{T_i}(\bar{r}_s+\lambda)du}\right]. \tag{21}$$

TRS PDE.

$$\frac{\partial V}{\partial t} + a\frac{\partial V}{\partial \lambda} + \tfrac{1}{2}b^2\frac{\partial^2 V}{\partial \lambda^2} - \lambda V - r_w W - rL + \lambda(B_0 - R) = 0. \tag{22}$$

Under full CSA ($L_t = V_t$, $W_t = 0$), the TRS NPV is

$$V(t) = \mathbb{E}_t^Q\Bigg[\int_t^T \lambda(B_0 - R)e^{-\int_t^{\tau}(r+\lambda)du}d\tau + (M_0 r_f T + B_0 - B_T)e^{-\int_t^T(r+\lambda)du} - \sum_{i:T_i\le T} c_i e^{-\int_t^{T_i}(r+\lambda)du}\Bigg]. \tag{23}$$

Closed form for full CSA. Define

$$\bar{D}(t,T) = \mathbb{E}_t^Q[e^{-\int_t^T(r+\lambda)du}], \qquad \gamma(t,T) = e^{\int_t^T(\bar{r}_s - r)du} - 1. \tag{24}$$

Then

$$\boxed{V(t) = M_0 r_f T\,\bar{D}(t,T) + B_0\left(\bar{D}(t,T) + \mathbb{E}_t^Q\left[\int_t^T \lambda e^{-\int_t^{\tau}(r+\lambda)du}d\tau\right]\right) - B_t - fva} \tag{25}$$

with the bond-flavored fva

$$fva = \gamma(t,T)B_t - \sum_{i:t<T_i\le T} c_i\bar{D}(t,T_i)\,\gamma(t,T) - \mathbb{E}_t^Q\left[\int_t^T \lambda R\,\gamma(\tau,T)e^{-\int_t^{\tau}(r+\lambda)du}d\tau\right]. \tag{26}$$

Bond forward. Setting $r_f = 0$, $M_0 = 0$ in the above and solving $V(t) = 0$ for F (with $B_0 \to F$):

$$\widetilde{D}(t,T) = \mathbb{E}_t^Q[e^{-\int_t^T(\bar{r}_s+\lambda)du}], \quad F(t,T) = \frac{B_t}{\widetilde{D}(t,T)} - \int_t^T \lambda\, e^{\int_\tau^T(\bar{r}_s+\lambda)du} d\tau - \sum_{i:T_i\le T} c_i \frac{\widetilde{D}(t,T_i)}{\widetilde{D}(t,T)}. \quad (27)$$

For Treasuries ($\lambda \equiv 0$) this reduces to the cost-of-carry

$$F(t,T) = B_t\, e^{\int_t^T \bar{r}_s\, du} - \sum_{i:T_i\le T} c_i\, e^{\int_{T_i}^T \bar{r}_s\, du}. \quad (28)$$

Multi-period TRS, deterministic rates. With deterministic r and $\bar{r}_s$, the integrals from (23) simplify (via $D(t,T)$ and the survival $q(t,T)$) using

$$dpv(t,T) \equiv \mathbb{E}_t^Q\left[\int_t^T \lambda e^{-\int_t^\tau(r+\lambda)du} d\tau\right] = -\int_t^T D(\tau,T)\, dq(\tau,T), \quad (29)$$

$$\mathbb{E}_t^Q\left[\int_t^T \lambda R\, \gamma(\tau,T)\, e^{-\int_t^\tau(r+\lambda)du} d\tau\right] = -R\int_t^T \gamma(\tau,T) D(\tau,T)\, dq(\tau,T). \quad (30)$$

The full multi-period TRS valuation is then

$$\boxed{V(t) = \sum_{j=0}\Big[(M_j r_{f,j}\Delta t_j + F_j - F_{j+1})D(t,t_{j+1}) - c_i D(t,T_i) + (F_j - R)\big(dpv(t,t_{j+1}) - dpv(t,t_j)\big)\Big]} \quad (31)$$

with $F_j = F(t,t_j)$. For fixed-loan TRS, $M_j = M_0$; for mark-to-market TRS where the funding notional resets to the bond price, $M_j = F_j$.

7 Numerical results from the paper (sanity benchmarks)

Table 1 (full CSA, no repo spread). $S_0 = 100$, $\sigma = 50\%$, $r = 10\%$, $T = 1$. Tree matches analytic to 12 digits; e.g. ATI 1Y 4-period: $V = -0.35977699$.

Table 2 (repo-style, $r_b = r_c = 2\%$, $r_s - r = 5\%$). ATI 1Y, one period. Analytic $V = -0.52327778$. Tree: 100 steps $\to -0.52349$, 250 steps $\to -0.52336$, 1000 steps $\to -0.52330$.

Table 3 (uncollateralized, asymmetric funding). $S = K = 100$, $\sigma = 0.5$, $q = 0$, $T = 1$, $r = 0.1$, $r_s - r = 2\%$, $r_f = 12.131\%$, $r_b = 12\%$, $r_c = 15\%$, $s_b = 1.7\%$, $s_c = 4.2\%$, $\mu_b = 0.3\%$, $\mu_c = 0.8\%$.
Results (no CSA): $V^* = \mp 0.5701$, $V_{\text{payer}} = -0.7577$, $V_{\text{receiver}} = 0.3509$. Bid/ask spread $\approx 0.44\%$. Repo-style margin compresses XVAs by $\sim 10\times$.

8 Validation Notes for the Pricing Library

8.1 Building blocks

1. **OIS discount** $D(t,T)$ from a bootstrapped OIS curve.
2. **Repo-financing factor** $e^{\int_t^T(r_s-r)du}$. Smoke test: $r_s = r \Rightarrow$ factor $= 1 \Rightarrow fva = 0$.
3. **Risky discount** $\bar{D}(t,T) = D(t,T)\, q(t,T)$ when r and λ are independent and deterministic. Cross-check via (24).
4. **Survival** $q(t,T)$ from a hazard curve. For piecewise constant λ, verify $q(t,T) = \exp\big(-\sum_i \lambda_i \Delta_i\big)$.

8.2 Equity TRS, full CSA

5. Reproduce (7): at $t = 0$ ATI with $r_s = r$, verify $V_0 = s_f(T - t_0)S_0 D(0,T)$ – the trapped funding-spread PV.
6. With $r_s > r$, verify $V_0 = s_f(T - t_0)S_0 D(0,T) - (e^{(r_s - r)T} - 1)S_0$. The fva exactly cancels the funding-spread PV when $s_f = (r_s - r)$ and $D(0,T) \approx e^{-rT}$ – this is how ATI swaps achieve zero NPV.
7. **Forward consistency.** Set $M_0 = 0$, $r_f = 0$, vary T; check $V_t = 0 \Leftrightarrow S_0 = S_t e^{(r_s - q)(T-t)}$.
8. **Multi-period.** Reproduce paper Table 1 entries – in particular the 4-period ATI case $V = -0.35977699$.

8.3 Repo-style margined TRS

9. Implement (9); ensure L_t uses the previous time-step's value when computed in a tree.
10. Symmetric-funding closed form (11): smoke test against MC under $r_b = r_c$. Reproduce paper Table 2 analytic $V = -0.52327778$.
11. **Tree convergence.** For a 1Y TRS, compare 100 / 250 / 500 / 1000 steps to analytic; convergence is $O(1/N)$ because L_t is set at the previous step.

8.4 Trinomial tree implementation

12. Probabilities (12): verify $p_u + p_m + p_d = 1$ and all are non-negative under the standard CFL-type bound $\sigma\sqrt{2\Delta t} > |r_s - q|\Delta t$.
13. Drift check: at each node, $\mathbb{E}[S_{t+\Delta t}/S_t] = e^{(r_s - q)\Delta t}$. (Critical – the tree drift is r_s, not r!)
14. Variance check: $\mathrm{Var}[\ln S_{t+\Delta t}/S_t] \approx \sigma^2 \Delta t$.
15. Switching discount r_e: at each node implement

$$r_e(\text{node}) = \mu\, r + (1-\mu)\big[r_b \mathbf{1}\{W \le 0\} + r_c \mathbf{1}\{W > 0\}\big],$$

with W taken from the value at the previous step (decoupling).

16. Recursive multi-reset: confirm that for a single period ($K = 1$, $\tau_0 = t_0$, $\tau_1 = T$) the recursion collapses to the standard one-period rollback.

8.5 Uncollateralized TRS and XVA decomposition

17. Compute V^* via OIS and V via switching r_e; verify $V^* - V = U$ matches (14) computed by direct integration on the tree.
18. Decomposition self-check: reproduce paper Table 3 – in particular $V^*_{\text{payer}} = -0.5701$, $V_{\text{payer}} = -0.7577$, with

$$V^*_{\text{payer}} - V_{\text{payer}} = -\,U_{\text{payer}} = -\,(cva - dva + cfa - dfa).$$

19. **Bond-as-a-derivative.** Treat a fixed-rate bond as a TRS limit (no funding leg, no resets, $V_t > 0$ throughout); verify that $V^* + cva + cfa = V_{\text{bond,market}}$ – the "bond price recovery" test that classical CVA/FVA frameworks fail (Morini–Prampolini 2011).
20. Symmetry (B's cva = C's dva): when both parties run the same model with swapped roles, verify the law of one price.
21. **Implied repo from ATI swap.** For $V = 0$ ATI, solve for s_f given r_s, or solve for r_s given s_f. From paper Table 3: fixing $r_f = 12.131\%$ gives implied repo spreads 1.26723% (payer) vs 1.65867% (receiver) – the ∼40 bp asymmetry is the bid/ask.

8.6 TRS on defaultable bond

22. Implement bond pricing (21); for deterministic λ verify $B(t) = \int_t^{T_b} c_i e^{-\int_t^{T_i}(\bar{r}_s+\lambda)du} + e^{-\int_t^{T_b}(\bar{r}_s+\lambda)du} + R\int_t^{T_b}\lambda(\tau)e^{-\int_t^{\tau}(\bar{r}_s+\lambda)du}d\tau$.
23. Closed form (25): validate via Monte Carlo under a Cox-process default with intensity λ independent of r.
24. **Treasury limit.** Set $\lambda = 0$, R irrelevant; check that $\bar{D} = D$, γ becomes $e^{\int_t^T(\bar{r}_s - r)du} - 1$, and (27) reduces to (28).
25. **Zero-haircut limit.** Set $h = 0$ so $\bar{r}_s = r_s$; verify the equity-TRS fva form (8) re-emerges from (26) when $\lambda \to 0$ and there are no coupons.
26. **Multi-period bond TRS.** Implement (31); test against the tree for a single underlying with λ small. The $dpv(t, t_{j+1}) - dpv(t, t_j)$ kernel should equal $-\int_{t_j}^{t_{j+1}} D(\tau, t_{j+1})dq(\tau, t_{j+1})$ – a numerical quadrature regression test.
27. **Bond forward.** Set $r_f = 0$, $M_0 = 0$ in (23) and solve $V(t) = 0$. Cross-check with the explicit forward (27); the two answers must match within MC error.

8.7 Edge cases and limits

28. $\sigma \to 0$: TRS becomes deterministic; tree, FK and analytic all collapse to the same number.
29. $T \to 0$: $V \to V_0$ trivially; XVAs $\to 0$.
30. $\mu \to 1$ (full CSA, asymmetric funding): all XVAs $\to 0$; $V = V^*$.
31. $\mu \to 0$ (no CSA): $L = 0$, $W = V^*$; verify the integrand of (14) reduces to $(r_w - r)V^* e^{-\int r_w du}$.
32. Symmetric default ($s_b = s_c$, $\mu_b = \mu_c$, V symmetric in sign): cva should equal dva and cfa equal dfa, so $U = 0$ – a clean regression test for the indicator-function logic.

Things to Double-Check Against the Paper

- **Sign convention.** The paper writes the TRS payoff from the *payer's* side; the bank B is the payer (lender of cash). If the library defaults to the receiver's perspective, flip the sign of $H(T)$ throughout.
- **Stub period.** Equation (2) contains accrued interest $M_0 r_f(t - t_0)$ on the funding leg – this is the *dirty* value. Library convention may differ.
- **Repo rate scope.** r_s is the *stock-specific* or *bond-specific* repo rate, not GC. For specials this can deviate materially from GC repo and should be a per-CUSIP input.
- **Funding-leg notional reset.** For mark-to-market TRS the funding notional resets to the prevailing bond price ($M_j = F_j$); for fixed-position/loan TRS $M_j = M_0$. This drives the multi-period formula (31).
- **All-in financing rate.** The bond case uses $\bar{r}_s = (1 - h)r_s + hr_N$ – the haircut portion is funded by B's rolled debt at r_N. This is the *lender* bank's all-in cost; mirror it for the borrower if the convention differs.
- **Sign in dva, dfa.** The minus sign in (16), (18) comes from the indicator $\{V < 0\}$ which makes the integrand sign-flip relative to cva/cfa – so when the paper writes $U = cva - dva + cfa - dfa$, the negative signs are already absorbed in the definitions.

Bibliographic Cross-References

The paper builds on a series of Lou's prior work on liability-side pricing and post-crisis discounting:

- Lou (2015) – foundational liability-side derivation, source of the spread decomposition $r - r = s + \mu$.
- Lou (2016a) – swap-specific liability-side framework.
- Lou (2016b) – KVA + repo gap risk; gives the analytical estimator for r_s when no liquid quote is available.
- Lou (2017) – discounting framework with non-cash collateral; basis for the switching-rate formulation (5).

The Piterbarg (2010) result on OIS discounting under full CSA is recovered as a special case ($\mu = 1 \Rightarrow r_e = r$); the Burgard–Kjaer (2011) framework is structurally similar but couples U to the exposure rather than to a switching effective rate.